EBRAR ACAR

Software Engineering Student

SUMMARY

Senior Software Engineering student specializing in autonomous multi-agent systems, swarm robotics, and edge AI. Experienced in developing mission software for multi-UAV swarm systems and multi-agent reinforcement learning based traffic signal control architectures, with strong practical skills in ROS, Python, and C++. Seeking to apply this experience to swarm autonomy and edge intelligence challenges in the space sector.

EDUCATION

Software Engineering **2023-Present**

Konya Technical University
GPA - 3.83

High School **2018-2023**

Mahmut Sami Ramazanoğlu AİHL
GPA - 3.92

- Improved English skills during preparatory class.
- Developed language and presentation skills by participating in symposiums, debate competitions, and Model United Nations conferences.

EXPERIENCE

Mosaş Group - Artificial Intelligence Intern **June - August 2026**

- Worked on multiple traffic signal control architectures using multi-agent reinforcement learning to coordinate intersections and reduce congestion.
- Tested these across different simulators and traffic networks, including a real intersection network, now being prepared as a research paper.

PROJECTS

Task Allocation and Autonomous Coordination for a ROS-Based UAV Swarm System - Capstone Project

- Developed mission software for a 5-UAV swarm system using ROS Noetic, including one carrier UAV.
- **Carrier UAV:** Developed a Brute Force based route planning software in C++ that guides 4 mini UAVs to their target zones in the shortest time.
- **Human Tracking:** Developed a stable tracking system combining a Python-based image processing node (MobileNet-SSD object detection, SORT/Kalman Filter based ID tracking, and uniform color based authorization) with C++-based PID control (forward-backward, lateral, yaw) and LIDAR integration.
- **Area Scanning:** Implemented autonomous area scanning with MobileNet-SSD based object detection in a Python image processing node, C++-based PID controlled scanning with user-defined parameters, and a safe payload drop after detection.
- **Object Detection:** Designed a system using OpenCV that performs area scanning on a fixed route with AruCo marker detection, transmitting detected coordinates in real time to the kamikaze UAV.
- **Kamikaze:** Developed autonomous flight software in C++ that simulates precise dive maneuvers toward target coordinates using PID control.
- The swarm coordination architecture is currently under development.

Multi-Agent Reinforcement Learning Based Traffic Signal Control Benchmark - Internship Project

- Tested multi-agent reinforcement learning (MARL) methods on the LibSignal framework, which supports cross-simulator testing, modeling intersections as coordinated agents that share neighboring intersection states to reach a common traffic flow goal, on a real 4×4 intersection network.
- Benchmarked CoLight and MPLight, which rely on inter-agent coordination, against Fixed-time and MaxPressure methods that use no information sharing, to measure coordination's contribution to reducing congestion.
- Repeated the same study across two different simulators (CityFlow and SUMO) and two different traffic distributions (homogeneous and heterogeneous vehicle composition) to examine the portability and consistency of coordination across different environments.
- Identified and fixed real bugs breaking coordination in the LibSignal implementations of MPLight and CoLight.
- Ported the DHLight model, originally CityFlow-only, to SUMO, integrated it into the LibSignal framework myself, and successfully tested it in the benchmark.

IoT-Based Fully Autonomous UAV Project - 2025 Teknofest International UAV Competition

- Developed ROS-based autonomous route calculation and payload drop software in Python.
- Set up a cloud-based MQTT communication infrastructure to establish a command network between Raspberry Pi units on different networks and the ground station computer.

rotAY - Hybrid Autonomous Route Optimization System for Lunar Rovers (TUA Astro Hackathon Project)

- Served as team lead in developing a hybrid autonomous route planning system based on A* and Genetic Algorithm, working on NASA LRO elevation data (MoonPlanBench).
- Designed and implemented a profile-based cost function that simultaneously optimizes distance, safety clearance, and terrain roughness, with three mission scenarios: Fast, Balanced, and Safe.
- Built a two-layer planning architecture: global waypoint optimization with Genetic Algorithm before the mission, and real-time local A* re-planning for dynamic obstacles.
- Developed the simulation environment and managed testing using Python, NumPy, SciPy, and scikit-image.

Turtlebot3 Cleaner

- Developed a QR-code-based cleaning robot software using ROS Noetic.
- Built an OpenCV-based ROS node for QR code detection.
- Performed environment mapping via RViz using SLAM algorithms.
- Implemented autonomous navigation using move_base.
- Designed the system with custom launch files for a modular, manageable structure.

TECHNICAL SKILLS

- **Programming Languages:** C++ • Python
- **Embedded Systems & Robotics:** Raspberry Pi • Jetson Nano • ROS • Gazebo
- **Communication & Networking:** TCP/IP • MQTT
- **Software Design:** SOLID Principles • Clean Code • OOP • Design Patterns
- **Tools & Other:** Docker • Git

LANGUAGES

- **English** - C1 (YÖKDİL: 90)

CONFERENCES

- **International Astronautical Congress 2026**

My research titled "Comparative Analysis of Single-Vehicle VIO vs. Swarm-Based Geometric Positioning for Mars Aerial Exploration Missions," comparing the performance of swarm aerial vehicles using geometric triangulation against single-vehicle visual-inertial odometry for Mars exploration, was accepted for an interactive presentation at IAC 2026.

CERTIFICATES & AWARDS

- **Defense Industry Academy 401 Training (Savunma Sanayii Akademi – 2026)**

A training program covering defense industry projects, systems engineering, and project management.

- **Autonomous Driving Technologies Program – Basic Training (Ministry of Industry and Technology – 2026)**

Technical expertise training focused on data processing and autonomous system integration, supported by leading companies in the sector.

- **3rd Place, Konya Technical University Project Market**

Our project "Autonomous Multi-UAV System V2," for which I developed the software component as part of my capstone project, won 3rd place at Konya Technical University's 10th Project Market.